

Experiment 15 :

Implement Probabilistic Context-Free Grammar (PCFG) Parsing for a Sentence Using Python

Code:

```
import nltk

from nltk import PCFG

from nltk.parse import ViterbiParser

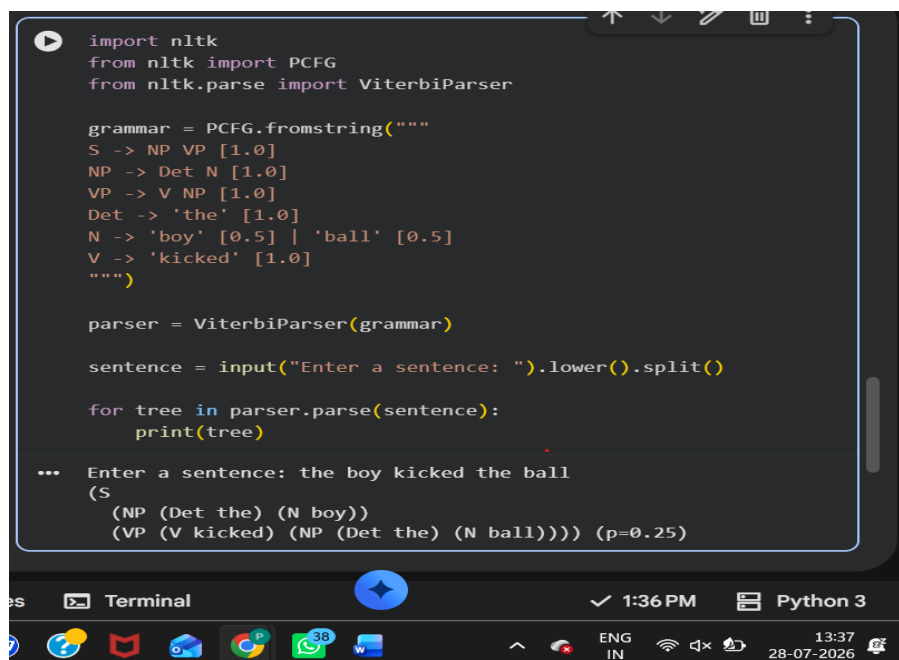
grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [1.0]
VP -> V NP [1.0]
Det -> 'the' [1.0]
N -> 'boy' [0.5] | 'ball' [0.5]
V -> 'kicked' [1.0]
""")

parser = ViterbiParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

for tree in parser.parse(sentence):
    print(tree)
```

Output



```
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for tree in parser.parse(sentence):
    print(tree)

... Enter a sentence: the boy kicked the ball
(S
  (NP (Det the) (N boy))
  (VP (V kicked) (NP (Det the) (N ball)))) (p=0.25)
```